Physics Question Bank
Questions + Answers
Unit 10

Knowledge covered:

الوحدات التدريبية	Unit 10: Electromagnetic induction	
أكواد المعارف	TPK03 – TPK04 - TPK05	
المعارف	TPK03	Meaning of Electromagnetic Induction and Faraday’s Law
	TPK04	Lenz’s Law
	TPK05	Mutual and self-induction The electric generator (dynamo) The electric transformer

Knowledge in each lesson:

Unit 10	Lesson 1	Lesson 2	Lesson 3	Lesson 4	Lesson 5
Knowledge code	TPK03	TPK04	TPK05	TPK05	TPK05

Questions expected in almost each lesson:

1. Write the scientific term expressed by the following statements.
2. Mention the factors affecting each of the following (one factor only)
3. Put circle around the correct answer
4. Correct the underlined in the following statement
5. True or False
6. Answer the following question
    i. Solve questions (calculate / find / determine)
    ii. Mention the rule or method used to determine
    iii. Draw the shape
    iv. State
    v. Write the mathematical law used to calculate
    vi. Give a reason for each of the following
7. Compare between each of the following

Lesson (1): Meaning of electromagnetic induction and faraday's law

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	phenomenon in which an induced electromotive force and an induced current are generated in the conductor by a changing magnetic field (magnetic flux). (دور أول 2024) (دور ثان 2023)	electromagnetic induction.
2	An electromotive force (current) is induced in a conductor while it is moving (cutting) across the magnetic field.	motional electromotive force (EMF) or flux cutting.

2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	the formed poles in solenoid by electromagnetic induction.	*Direction of Current *Direction of Magnetic Field *Relative Orientation of Solenoid and Magnetic Field *Number of Turns in the Solenoid *Core Material
2	the direction of the induced emf by electromagnetic induction.	*Direction of the magnetic field *Direction of motion of the conductor *Rate of change of magnetic flux
3	emf induced by electromagnetic induction (faraday's experiment).	*rate of change of magnetic flux *Magnetic Field Strength *Area of the Conductor *Relative Motion between Conductor and Magnetic Field *Angle between Conductor and Magnetic Field *Number of Turns in a Coil

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	A moving conductor coil produces an induced e.m.f. This is in accordance with	a) Lenz's law b) Faraday's law c) Coulomb's law	b
2	Weber per square meter is a) henry b) ampere c) Tesla d) Ohm		c
3	Which of the following statements is true about an electromagnet? a) An electromagnet is permanent magnet b) The strength of an electromagnet can be changed by changing the number of turns in its coil		b

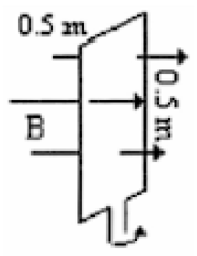
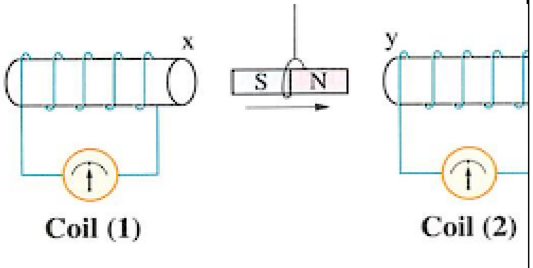
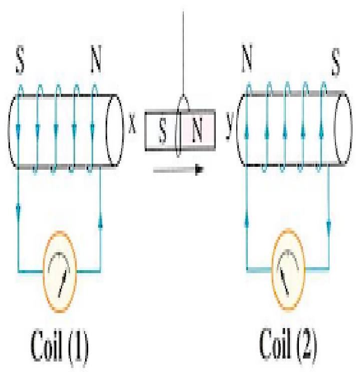
	c) An electromagnet produces a comparatively weak force of attraction d) The polarity of an electromagnet cannot be changed	
4	If a coil is placed in a uniform magnetic field, then a) No induced emf generated. b) an induced current flow in the coil. c) an emf is induced in the coil. d) None of the above	a
5	A coil of copper wire is moved parallel to a uniform magnetic field. The e.m.f. induced in the coil will: a) depend on the intensity of magnetic field b) depend on the velocity of the coil c) be zero d) be infinite	c

4. Correct the underlined in the following statement

No.	Question	Answer
1	The process of producing electric current through the variable magnetic field is called Faraday's Law	Electromagnetic Induction
2	According to Faraday's law of electromagnetic induction, electromotive force is produced by time varying magnetic force.	Flux
3	If the speed by which a magnet approaching a solenoid is constant, then the induced e.m.f. will be zero	Zero
4	The polarity of induced e.m.f. is given by Faraday's law	Lenz`s law
5	The magnitude of the induced e.m.f. in a conductor depends on the speed of change of the magnetic flux within the conductor	Rate

5. Answer the following question

No.	Question	Answer
1	A coil of an area 2 m^2 is placed in a magnetic field which changes from 4 Wb/m2 to 8 Wb/m^2 in 2 seconds. Find the induced e.m.f. in the coil	4V
2	A square loop is placed in a uniform magnetic field perpendicular to the plane of the loop as shown. The loop is 0.50 meters on a side and the magnetic field B has a strength of 2 T. If the loop is rotated through an angle of 90° in 0.1 second what	5V

	would be the average induced EMF in the loop? 	
3	A square coil having 500 loops of side 10 cm is placed normal to magnetic flux which increases at a rate of 1 T/s. The induced emf is	$EMF = -N\frac{\Delta\emptyset}{\Delta t}$ $EMF = -N\frac{A\Delta B}{\Delta t}$ $EMF = -500 * \left(10 * 10^{-2} * 10 * 10^{-2}\right) * 1$ EMF = -5 V
4	following figure shows a magnet placed between two coils, each of them is connected to a moving coil galvanometer whose zero reading is at the middle of its scale. If the magnet is moved in the shown direction as in the figure:  a) Determine on the drawing the direction of the induced current generated in the two coils. b) Determine the type of the magnetic poles which are formed at x. y.	
5	A coil of 200 turns intercepts magnetic flux of 7x 10^{-3} Wb. If this magnetic flux density has changed uniformly to 5x10^{-3} Wb within 0.1 s, calculate the induced electromotive force in the coil	$\Delta\phi_m = (\phi_m)_2 - (\phi_m)_1 = (5 \times 10^{-3}) - (7 \times 10^{-3}) = -2 \times 10^{-3}$ Wb $emf = -N\frac{\Delta\phi_m}{\Delta t} = -200 \times \frac{-2 \times 10^{-3}}{0.1} = 4\,V$

6	A square coil of side length 10 cm and 500 turns is placed perpendicular to a uniform magnetic field of flux density 0.1 T. Calculate the induced emf in the coil, if a) The magnetic flux fades within 0.15 s. b) The flux density is increased to 0.3 T within 0.75 s	a) $\text{emf} = -N\frac{(-BA)}{\Delta t} = \frac{-500\times10^{-2}\times(-0.1)}{0.15} = 3.33\text{ V}$ b) $\text{emf} = -NA\frac{\Delta B}{\Delta t} = \frac{-500\times10^{-2}\times(0.3-0.1)}{0.75} = -1.33\text{ V}$
7	The magnetic flux through a 50-turn coil increases at the rate of 0.05 Wb/s. What is the induced emf between the ends of the coil?	Given that N=50 turns $\frac{\Delta\phi}{\Delta t} = 0.05$ Wb/s induced emf $= -N\frac{\Delta\phi}{\Delta t}$ $= 50 \times 0.05 =$ **2.5 Volt**
8	The magnetic flux in a coil change from 20 × 10^{-2} Weber to 12 × 10^{-2} Weber in 0.02 s. Find the value of induced emf in the coil.	$e = -\frac{\Delta\phi}{\Delta t} = -\frac{(12\times10^{-2}-20\times10^{-2})}{0.02} = 4\,V$
9	A coil of an area 0.2 m^2 is placed in a magnetic field which changes from 5 Wb/m^2 to 10 Wb/m^2 in 0.2 seconds. Find the induced e.m.f. in the coil	$A = 0.2\,m^2$ $B_1 = 5\,Wb/m^2$ $B_2 = 10\,Wb/m^2$ $\Delta t = 0.2\,s$ $\phi = BA$ $\phi_1 = B_1A = 5\times0.2 = 1\,Wb$ $\phi_2 = B_2A = 10\times0.2 = 2\,Wb$ $Emf = -N\frac{\Delta\phi}{\Delta t}$ ولأنه لا توجد معلومات عن عدد اللفات $N=1$ $Emf = -1\frac{\phi_2-\phi_1}{\Delta t} = -\frac{2-1}{0.2}$ $Emf = \boxed{-5\,V}$ اذا كانت من ضمن الاختيارات بدون إشارة السالب فلا مشكلة

10	Calculate the magnitude of the induced emf when the magnet is thrust into the coil, given the following information: the single loop coil has a radius of 6.00 cm and the average value of B cosθ (this is given, since the bar magnet's field is complex) increases from 0.0500 T to 0.250 T in 0.100	We are given that $N = 1$ and $\Delta t = 0.100$ s, but we must determin the change in flux $\Delta\phi$ before we can find emf. Since the area of the loop is fixed, we see that $$\Delta\phi = \Delta(BA\cos\theta) = A\Delta(B\cos\theta)$$ Now $\Delta(B\cos\theta) = 0.200$ T, since it was given that $B\cos\theta$ changes from 0.0500 to 0.250 T. The area of the loop is $A = \pi r^2 = (3.14\ldots)(0.060 \text{ m})^2 = 1.13 \times 10^{-2} \text{ m}^2$. Thus, $$\Delta\phi = (1.13 \times 10^{-2} \text{ m}^2)(0.200 \text{ T}).$$ Entering the determined values into the expression for emf gives $$\text{Emf} = N\frac{\Delta\phi}{\Delta t} = \frac{(1.13 \times 10^{-2} \text{ m}^2)(0.200 \text{ T})}{0.100 \text{ s}} = 22.6 \text{ mV}$$
11	A coil having 1000 loops of side length 20 cm is placed normal to a magnetic flux which increases at a rate of 0.5 T/s. then induced e.m.f. is	$N = 1000$ $S = 20$ cm $A = s^2 = (0.2)^2 = 0.04 \text{ m}^2$ $\frac{\Delta B}{\Delta t} = 0.5$ T/s As $\phi = BA$ $\frac{\Delta\phi}{\Delta t} = \frac{\Delta B}{\Delta t} A$ $\frac{\Delta\phi}{\Delta t} = 0.5 \times 0.04 = 0.02$ Wb/s $E.m.f = -N\frac{\Delta\phi}{\Delta t}$ $= -1000\ (0.02)$ $E.m.f = -20$ V
12	A coil of area 50 cm^2 has 1000 turns. Magnetic field of 0.2 weber/m^2 is perpendicular to the coil. The magnetic field is reduced to zero in 0.2 second. The induced emf in the coil is	**CALCULATION:** Given θ = 0, A = 50 cm^2, N = 1000, B_1 = 0.2 weber/m^2, B_2 = 0 weber/m^2, and dt ⇒ A = 50 cm^2 = 50 × 10^{-4} m^2 • The change in the magnetic field is given as, ⇒ dB = B_2 - B_1 ⇒ dB = 0 - 0.2 ⇒ dB = - 0.2 weber/m^2

		$\Rightarrow d\phi = A.dB.\cos 0$ $\Rightarrow d\phi = A.dB$ $\Rightarrow d\phi = -50 \times 10^{-4} \times 0.2$ $\Rightarrow d\phi = -10 \times 10^{-4}$ $\Rightarrow d\phi = 10^{-3}$ weber -----(1) • By Faraday's law of electromagnetic induction, the magnitude of the induced emf i given as, $\Rightarrow e = -N\frac{d\phi}{dt}$ $\Rightarrow e = -1000 \times \frac{-10^{-3}}{0.2}$ $\Rightarrow e = 5$ V
13	A coil of 800 turns, each of area 36 cm^2, the coil is affected by a normal magnetic flux relative to its plane, the flux density is $5.3*10^{-6}$T, if the magnetic flux is vanished in 1.5 ms. Calculate the induced EMF.	*Givens:* $N = 800$ $A = 36cm^2 = \frac{36}{100*100}m = 3.6*10^{-3}m$ $B = 5.3*10^{-6}T$ $t = 1.5ms = 1.5*10^{-3}s$ $\boxed{EMF_{ind} = \frac{-N\,\Delta\Phi_m}{\Delta t}}$ $EMF = -\frac{800*(3.6*10^{-3}*(0-5.3*10^{-6})}{1.5*10^{-3}}$ $EMF = 0.010175V$
14	Mention the application of electromagnetic induction.	Electrical Generators(dynamos) Transformers
15	Mention the scientific base electric generators.	Electromagnetic induction.
16	A square loop of 10 turns is placed in a uniform magnetic field perpendicular to the plane of the loop, if the magnetic flux changes from 0.03 Wb to zero in 0.1 s. Calculate the average induced emf in the loop. (تجريبي)	emf = -N * ($\Delta\Phi$ / Δt) $\Delta\Phi = \Phi_2 - \Phi_1 = 0 - 0.03 = -0.03$ Wb emf= -10 * (-0.03 / 0.1) = 3 V the average induced EMF in the loop is 3 volts

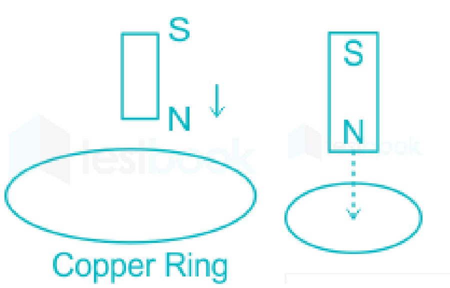
17	A metallic ring is placed in a uniform magnetic field perpendicular to the plane of the ring. If the magnetic flux changes from 0.05 Wb to – 0.05 in 0.05 s. Calculate the average induced EMF in the loop. (دور أول 2023)	emf= -N * (ΔΦ / Δt) we're dealing with a single ring, the number of turns, N, is 1. $\Delta\Phi = \Phi_2 - \Phi_1 = -0.05 - 0.05 = -0.1$ Wb emf = -1 * (-0.1 / 0.05) = 2 V the average induced EMF in the loop is 2 volts.

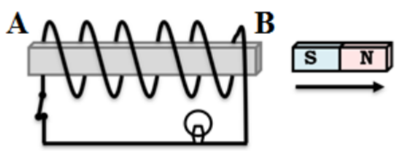
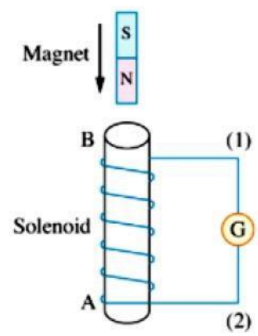
Lesson (2): Lenz's law

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	The negative sign in faraday's relation that the direction of the induced electromotive force or the induced current tends to oppose the cause producing it.	Lenz's Law
2	The induced current must be in a direction such as to oppose the change producing it. (تجريبي)	Lenz's Law

2. Put circle around the correct answer

No.	Question	Choices	Answer
1	The induced current must be in a direction such as to oppose the change producing it	a) Electromagnetic Induction b) Lenz's law c) Faraday's Law	b
2	A bar magnet is dropped and it passes through a copper ring as shown in the diagram. The acceleration of the falling magnet while passing through the ring is 	a) equal to acceleration due to gravity Faraday b) more than the acceleration due to gravity c) less than the acceleration due to gravity d) Zero	c
3	Identify the law used to find the direction of eddy current	a) Lenz's law b) Faraday's Law	a
4	When a conductor connected to an electrical circuit moves in a magnetic field, the direction induced current depends upon	a) number of turns and length of the conductor b) strength of the magnetic field c) speed of movement of the conductor d) direction of the magnetic field and direction of motion of the conductor	d
5	When the magnet is plugged out of the coil, the induced current in the coil will experience	a) Attraction force b) Repulsion force c) Need more givens	a

<table>
<tr><td>6</td><td>When the magnet is plugged into the coil, the induced current in the coil will experience</td><td>a) Attraction force
b) Repulsion force
c) Need more givens</td><td>b</td></tr>
<tr><td>7</td><td>A permanent magnet is quickly introduced into the center of an aluminum closed ring. With this ring ...</td><td>a) will be attracted to the magnet
b) push away from the magnet
c) remain stationary</td><td>b</td></tr>
<tr><td>8</td><td>In the opposite figure, when the magnet moves in the direction shown the pole formed at A is …(2023 دور ثان)

</td><td>a) North
b) South
c) West
d) east</td><td>a</td></tr>
<tr><td>9</td><td colspan="2">A bar magnet falls towards a solenoid as in the opposite figure, then during the approaching of the magnet towards the coil, which of the following choices?
<table>
<tr><th></th><th>The direction of the current in the galvanometer</th><th>The type of the formed pole at (A)</th></tr>
<tr><td>(A)</td><td>From (2) to (1)</td><td>north</td></tr>
<tr><td>(B)</td><td>From (1) to (2)</td><td>south</td></tr>
<tr><td>(C)</td><td>From (2) to (1)</td><td>south</td></tr>
<tr><td>(D)</td><td>From (1) to (2)</td><td>north</td></tr>
</table>

</td><td>c</td></tr>
<tr><td>10</td><td colspan="2">According to the Lenz's law…………(2024 دور أول)
a) The direction of the induced emf will be such that it helps the change in magnetic flux
b) The direction of the induced current will be such that it opposes the change in magnetic flux.
c) The magnitude of the induced electromotive force (emf) is directly proportional to the rate of change of flux.
d) The direction of the induced electromotive force (emf) will be along the direction of the magnetic field.</td><td>b</td></tr>
</table>

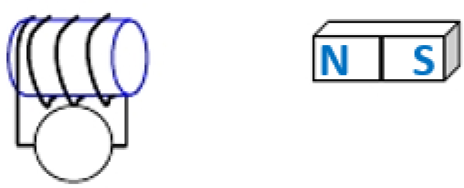
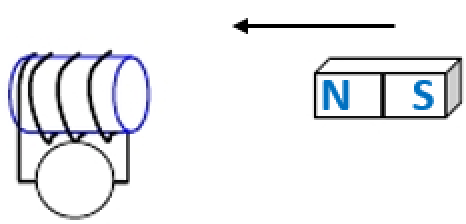
3. Correct the underlined in the following statement

No.	Question	Answer
1	The coefficient of mutual inductance between two coils increases as the rate of change of current in the primary coil decreases. (2023 دور ثان)	Decreases
2	Lenz's law is used to determine the magnitude of emf induced in the coil	Direction

4. True or False

No.	Question	Answer
1	Lenz's law is used to determine the magnitude of emf induced in the coil	False
2	Induced current will appear in such a direction that it opposes the change that produced it	True
3	The Lenz's law is consistent with the law of conservation of energy	True
4	When the current flowing through a wire reverses direction, the magnetic field around the wire reverse direction	True
5	If the current in a wire double, the induced magnetic field remain the same	False
6	Magnetic flux is maximum when the plane of the loop is parallel to the magnetic field	False

5. Answer the following question

No.	Question	Answer
1	Mention the application of Lenz's rule	
2	In the opposite figure: What is the direction of the induced EMF? a)  b)  c)	

6. Compare between each of the following

<table>
<tr><th>No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1</td><td colspan="2">Compare (2023 دور أول)
<table>
<tr><td>Point of comparison</td><td>Lenz’s rule</td><td>The right cork screw rule</td></tr>
<tr><td>The use</td><td>Lenz's law is used to determine the magnitude of emf induced in the coil</td><td>used to determine the direction of the magnetic field produced by an electric current.</td></tr>
</table>
</td></tr>
</table>

Lesson (3): Mutual and self-induction

1. Write the scientific term expressed by the following statements.

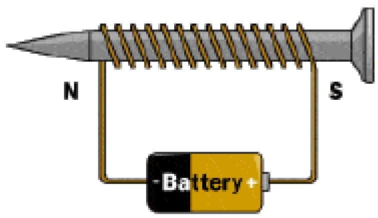
No.	Question	Answer
1	It's the electromagnetic effect takes place between two coils when an induced emf generated in one of them (secondary coil) due current variation in the other coil (primary coil), which opposes the change causing it. (تجريبي)	mutual inductance (mutual induction)
2	The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil. (دور أول 2023)	self-induction.

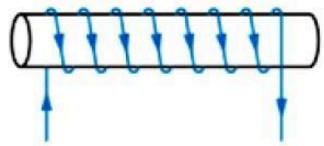
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	emf induced by mutual induction	1. Number of Turns in the Primary Coil: 2. Number of Turns in the Secondary Coil: 3. Distance between the Two Coils 4. Relative Orientation of the Coils 5. Permeability of the Core Material 6. Rate of Change of Current in the Primary Coil: 7. Frequency of the Alternating Current in the Primary Coil:
2	emf induced by self-induction	1-Rate of change of current 2-Self-inductance of the circuit
3	The coefficient of mutual inductance between two coils. (دور أول 2024) (دور أول 2023)	a) The presence of an iron core inside the coil. b) The volume of the coil and the number of its turns. c) The distance separating them.
4	The self-inductance (coefficient of self-induction)	1) The geometry of the coil (size, length and the number of turns) 2) The distance between the turns 3) The presence of an iron core inside the coil (magnetic permeability)

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	A transformer is based on the principle of	a) Mutual-inductance b) Self-inductance c) Both a and b	a
2	Changing current in a coil produces emf in the same coil. this phenomenon is known as	a) Mutual-inductance b) Self-inductance	b

		c) Both a and b	
3	The phenomenon due to which there is an induced current in one coil due to the current in a neighboring coil is?	a) Mutual-inductance b) Self-inductance c) Both a and b	a
4	If you wrap more coils around an electromagnet, it will become...  	a) Brighter b) Weaker c) Invisible d) Stronger	d
5	The production of electric current by a changing magnetic field is called	a) induced current b) electromotive force c) electromagnetic induction	c
6	The effect on inductance by adding extra coil layers	a) increase L b) decrease L c) Nothing happens d) Need more givens	a
7	The production of a voltage across an inductor as a result of its own expanding or contracting magnetic field is called	a) Mutual-inductance b) Self-inductance c) Both a and b	b
8	Henery is the unit of the a) Coefficient of mutual inductance only b) Coefficient of self-inductance only c) Both a and b d) None		c
9	When do we have self-induction? a) When the magnet moves into the coil b) When the coil moves into the magnet c) When a constant current passes through the coil d) When a changing current passes through the coil		d
10	A coil which carries current I is wound on iron core. The self-induced electromotive force in the coil is not affected by……… (2024 دور أول) a) Variation in coil current b) Variation in voltage to the coil c) Change of number of turns of coil d) The resistance of coil wire		d

11	In the opposite figure, when the iron rod is withdrawn from the coil, the self inductance of the coil ………..  (A) becomes zero. (B) decreases. (C) increases. (D) remains constant.	b
12	When the electric intensity passing through a coil of self-inductance 0.014H changes from 10A to zero, an induced emf of 7V is generated between the terminals of the coil. So, the time of the change in the electric current intensity is ………………. (A) 0.02s (B) 0.01s (C) 0.04s (D) 0.03s	a
13	The coil of self-induction 0.05H consisting of 100turns carries an electric current that produces a magnetic flux of 9×10^{-4}Wb through the coil. If the passing current in the coil has vanished through an interval of 0.03s, then the average induced emf in the coil equals ………….. and the intensity of the current which was flowing in the coil before it vanishes equals ………… (A) 3V, 3.6A (B) 3V, 1.8A (C) 12V, 3.6A (D) 12V, 1.8A	a

14	A solenoid of 500 turns has a self-inductance of 0.5H. If the intensity of the current that passes through the solenoid changes by (ΔI) in amperes, then the value of the change in the magnetic flux that cuts the solenoid during the same time is numerically equal to …. (A) $(0.01\Delta I)$Wb (B) $(10\Delta I)$Wb (C) $(50\Delta I)$Wb (D) $(0.001\Delta I)$Wb	b

4. Correct the underlined in the following statement

No.	Question	Answer
1	The induced electromotive force (emf) generated in a coil increases to <u>double</u> if the rate of change of current in the same coil increases 4 times. (2024 دور أول)	4 times
2	The mutual induction coefficient measured in <u>Ampere/second</u>. (دور أول 2023)	henry (H)

5. True or False

No.	Question	Answer
1	Switching the current in the primary coil on and off will produce an induced electromotive force in the secondary coil.	True
2	The mutual-inductance between two coil depends on number of turns in each of the two coils.	True
3	The mutual-inductance between two coil depends on the medium inside the coils	True
4	The self-inductance of a coil depends on the medium inside the coil.	True
5	The self-inductance of a coil depends on the current flowing through the turns of the coil.	False
6	The self-inductance of a coil depends on the separation between the turns of the coil	True
7	Self-induced emf is always proportional to the time rate of change of the current.	True
8	When a constant current of 5.0 A flows in a 400 turns solenoid of length, l and cross-sectional area A, emf is induced in the coil.	False

9	Self-inductance is vector	False
10	the voltage induced across an inductor is inversely proportional to the rate of change of magnetic flux.	False

6. Answer the following question

No.	Question	Answer
1	When current in a coil change from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self-inductance of the coil is	1.67 H
2	Determine the inductance of the coil if a 5 V self-induction EMF occurs at a rate of change of current of 10 A / s.	0.5 H
3	When the current in a coil is reduced uniformly from 20 A to zero in 0.50 s, an e.m.f. of 4.0 V is induced in a neighboring coil. What is the value of the mutual inductance between the two coils?	0.1 H
4	An application of mutual induction is	electric transformer
5	An application of self-induction	neon lamp
6	When current in a coil change from 6 A to 1 A in 0.2 seconds, an average voltage of 40 V is produced. The self-inductance of the coil is?	$L = (V * t) / \Delta I$ $L = (40\ V * 0.2\ s) / (6\ A - 1\ A)$ $L = 8\ V*s / 5\ A$ $L = 1.6\ H$
7	When a current of 2mA is supplied to a coil with 100 turns, a magnetic flux of magnitude 0.2Wb is linked with it. Find the self-inductance of this coil.	Current supplied to the coil I=2mA Number of turns in the coil (N)=100 Magnetic flux linked with the coil (ϕ)=0.2Wb Self-inductance of a coil is given by the equation $L=N\phi/I$ Substituting the values $L=100\times0.2/2$ This coil has a self-inductance of 10H
8	What is the principle of mutual induction?	Mutual induction is based on the electromagnetic induction concept. When one coil's magnetic field connects to the other, the second coil generates its own emf. Mutual induction is the term for this procedure.

9	A long solenoid has 500 turns. When a current of 2 A is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. find self-inductance.	$\Phi=4\times10^{-3}$ Wb/turn N= Number of turns =500 I= Current flowing =2A $N\times\Phi=L\times I$ $\therefore 500\times4\times10^{-3}$ $=L\times2$ $\therefore L= 500\times4\times10^{-3}/2$ $\therefore L=1H$ ∴ Self-inductance of solenoid is 1.0H
10	When a current of 4A between two coils changes to 12A in0.5s in primary and induces an emf of 50mV in the secondary. Calculate the mutual inductance between the two coils.	$M= e_2/ di_2/dt$ $=(50 \times 10^{-3}) / (8/0.5)$ $=3.125 \times 10^{-3}$ H =3.125 mH
11	Mention the application of 1- mutual induction. 2- self-induction. 3- the coil connected to neon lamp.	
12	Mention the scientific base 1- electric transformer. 2- neon lamp.	1- Faraday's Law of Electromagnetic Induction
13	An electric current of intensity 5 A flowing through a coil of self-inductance 0.001 H. If the current vanishes in 0.5 second, calculate the induced electromotive force (emf) generated in the coil. (2023 دور ثان)	emf = -L * (dI/dt) emf = -0.001 H * (-5 A / 0.5 s) = 0.01 V
14	when the current changes from +2A to -2A in 0.05 sec, an emf induced in a coil. Calculate the coefficient of self-induction in the coil. (تجريبي)	emf = -L * (dI/dt) dI = -2A - 2A = -4A dt = 0.05 sec EMF = 8V (assuming from the given information) 8V = -L * (-4A / 0.05 sec) 8V = L * 80 A/sec L = 8V / 80 A/sec = 0.1 H
15	When the current in a coil change from 5A to 2A in 0.05 s, an e m f of 15 V is induced	EMF = -L * (dI/dt) dI/dt = (2 A - 5 A) / 0.05 s = -60 A/s

	in the coil. Calculate the coefficient of self-induction of the coil. (2023 دور أول)	$15\ V = -L * (-60\ A/s)$ $L = 15\ V / 60\ A/s = 0.25\ H$

7. Compare between each of the following

<table>
<tr><th>No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1</td><td colspan="2">Compare
<table>
<tr><td>Point of comparison</td><td>Backward induced EMF</td><td>Forward induced EMF</td></tr>
<tr><td>Movement</td><td>Moving the primary coil near to or inside secondary coil</td><td>Moving the primary coil away from (take out) secondary coil</td></tr>
<tr><td>Using rheostat</td><td>Using rheostat, increase the intensity of the current in the primary coil.</td><td>Using rheostat, decrease the intensity of the current in the primary coil.</td></tr>
<tr><td>Using switch</td><td>Using switch, switch-on the primary circuit</td><td>Using switch, switch-off the primary circuit</td></tr>
</table></td></tr>
</table>

Lesson (4): The electric generator (dynamo)

1. Write the scientific term expressed by the following statements.

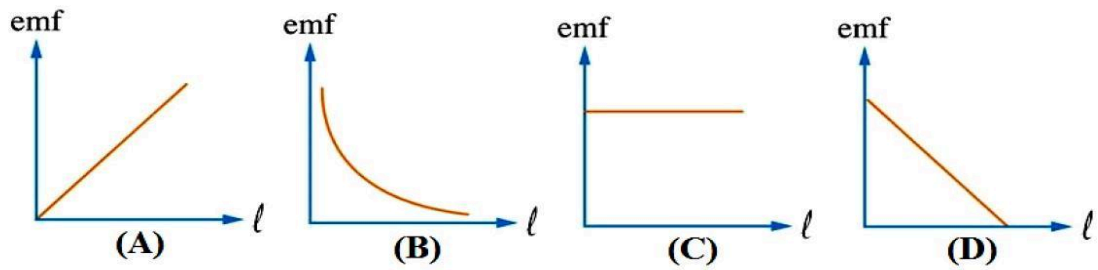
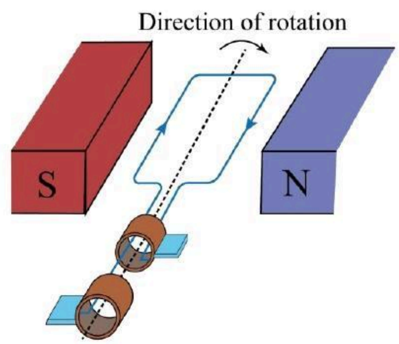
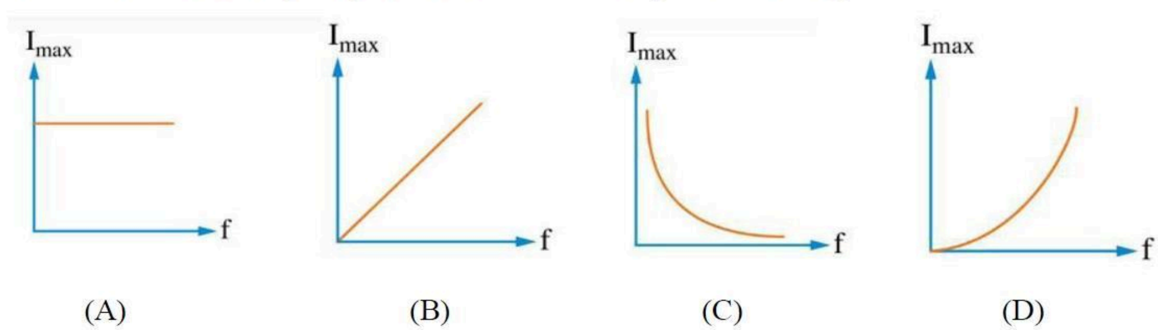
No.	Question	Answer
1	They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. (تجريبي) (دور أول 2023)	eddy currents

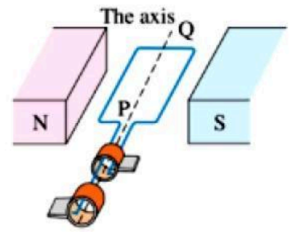
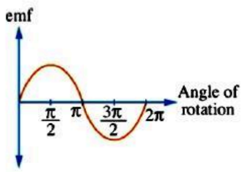
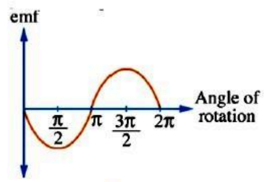
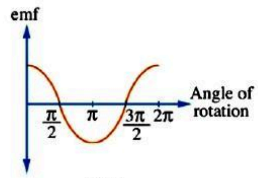
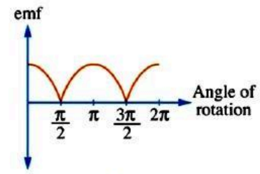
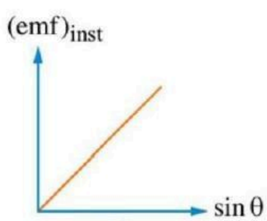
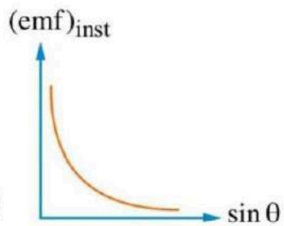
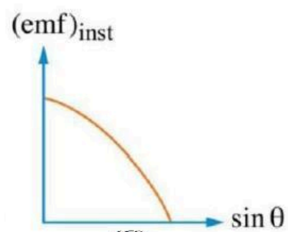
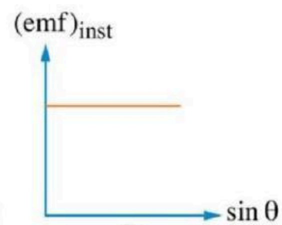
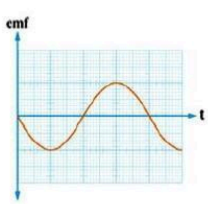
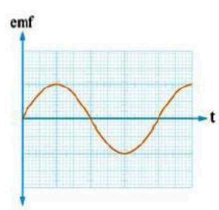
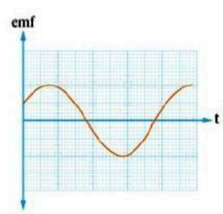
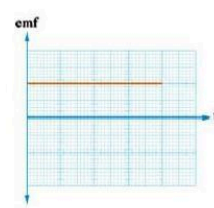
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	The induced emf in a straight wire moving normal to magnetic field. (تجريبي) (دور أول 2023)	1-Magnetic field strength (B) 2-Velocity of the wire (v) 3-Length of the wire (l) 4- Angle between the wire and the magnetic field
2	The direction of induced emf in a straight wire moving normal to a magnetic field	1- Direction of motion 2- Direction of the magnetic field 3- Orientation of the wire
3	The instantaneous value of induced emf generated in the dynamo. (دور أول 2024)	1-Strength of the magnetic field: 2. Number of turns in the coil 3- Area of the coil 4-Speed of rotation 5-Angle between the coil and the magnetic field 6-Shape of the coil

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	When a coil rotates in a magnetic field, the direction of the induced electromotive force produced in the coil is changed each ……………cycle (دور ثان 2023)	a) 1/4 b) 1/2 c) 1/1 d) 3/4	B
2	When the number of turns in the dynamo's coil is doubled at constant other factors then the maximum electromotive force generated from it…………… (دور أول 2023) (دور ثان 2023)	a) Increases to the double b) Decreases to half of its value c) remains constant d) Increases three times of its value	A

3	Which of the following graphs represents the relation between the induced electromotive force (emf) between the terminals of a wire that has a changeable length and the length of the wire (L), when it moves each time with the same uniform velocity perpendicular to a uniform magnetic field?  	a
4	The opposite figure represents an AC dynamo, if the output voltage was 10V when the dynamo's position was an as in the figure, so it becomes (-10V) when the coil rotates by an angle of …… (A) 270° (B) 180° (C) 90° (D) 360°  **Answer:**	b
5	Which of the following graphs represents the relation between the maximum value of the AC current (I_{max}) which is produced from a dynamo connected to an ohmic resistance and the frequency of the rotation of the dynamo's coil (f)?  	b

6	The opposite figure shows a rectangular coil that rotates between the poles of a magnet. If the coil rotates about its axis (PQ) starting from the position shown in the opposite figure, so which of the following graphs represents the change of the induced emf in the coil through one complete cycle?   (A)  (B)  (C)  (D)	c
7	Which of the following graphs represents the relation between the magnitude of the instantaneous induced electromotive force $(emf)_{inst}$ in the dynamo's coil and the sine of the angle of rotation of the coil ($\sin\theta$) starting from zero position?  (A)  (B)  (C)  (D)	a
8	which of the following graphs represents the relation between the induced instantaneous electromotive force and time in a simple dynamo, if the coil started rotating from the position where the plane of the coil was inclined to the field by an angle of 60°?  (A)  (B)  (C)  (D)	C

9	The magnitude of the instantaneous induced electromotive force which is generated in the dynamo's coil when the magnetic flux that penetrates it is the maximum value equals .... (A) the effective value. (B) zero. (C) the maximum value. (D) the average value.	B
10	which of the following will cause an induced current in a coil of wire (تجريبي) a) A magnet resting near the coil. b) the constant field of the earth passing through the coil c) A magnet being moved into or out of the coil. d) A wire carrying a constant current near the coil.	C
11	A straight wire 1m long is moving in a uniform magnetic field of flux density 0.1T at a uniform velocity 10m/s. an induced potential difference of 0.5V is generated between its terminals. The angle between the direction of the wire motion and the direction of the magnetic field is. (2024 دور أول) a) 0° b) 30° c) 60° d) 90°	D
12	A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is parallel to the magnetic field is (100 V). If the frequency of rotation is doubled, then the instantaneous value of emf produced at the same position is………… (2023 دور أول) (a) 50V (b) 100V (c) 200V (d) 400V	B

4. Correct the underlined in the following statement

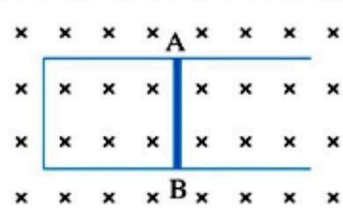
No.	Question	Answer
1	The ac generator is a device based on the principle of magnetic torque. (دور أول 2024)	Electromagnetic induction.
2	The electromotive force induced in a rectangular coil rotating in a magnetic field becomes zero when the coils' plane is parallel to magnetic field. (دور ثان 2023)	perpendicular
3	Maximum induced emf in a straight wire moving parallel to a magnetic field. (2023 دور ثان)	perpendicular

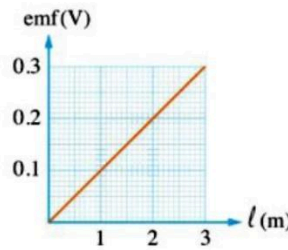
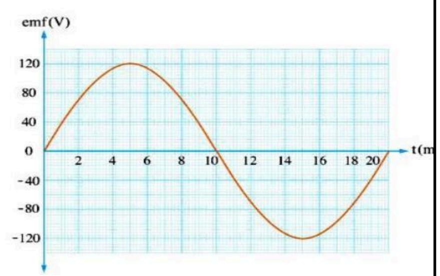
5. True or False

No.	Question	Answer

1	A straight wire moving parallel to a magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is closed induced current will flow in a direction identified by Fleming Right Hand Rule	False
2	A straight wire moving in a perpendicular magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is closed induced current will flow in a direction identified by Fleming Right Hand Rule	True
3	A straight wire moving in a perpendicular magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is open induced current will flow in a direction identified by Fleming Right Hand Rule	False
4	AC currents are produced by generators and batteries	False

6. Answer the following question

No.	Question	Answer
1	- A straight wire 0.5m long is moving in a uniform magnetic field of flux density 0.2T at a uniform velocity 10 m/s. An induced potential difference of 0.5 V is generated between its terminals. Find the angle between the direction of the wire motion and the direction of the magnetic field.	**Solution:** EMF_{ind} **= B L V sin (θ)** **(0.5) = (0.2)(0.5)(10) sin (θ)** **sin (θ) = 1/2** **Θ =** 30^{o}
2	The opposite figure shows a metallic wire (AB) of length 0.15m that is placed perpendicular to a magnetic flux of density 0.4T. When the wire is moved in the magnetic flux with a uniform velocity (v) in a certain direction, an induced emf of 0.03V is generated between its terminals which leads to the flow of an electric current from terminal (A) to terminal (B) across the wire, then the speed of the wire is …… and the direction of the wire velocity is …… (A) 1m/s, to the left of the page. (B) 1m/s, to the right of the page. (C) 0.5m/s, to the left of the page. (D) 0.5m/s, to the right of the page. 	Solution EMF_{ind} = B L V sin (θ) (0.03) = (0.4)(0.15)(V) sin (90) (v) = (0.03) / (0.4) (0.15) (v) = 0.5 m/sec **To the left**

3	The opposite graph represents the relation between the induced electromotive force in straight wire that moves with a uniform velocity (v) perpendicular to a magnetic flux density 0.1T and the length of that wire, so the magnitude of its velocity is .... (A) 1m/s (B) 2m/s (C) 0.2m/s (D) 0.1m/s  **Answer:** **(A) 1m/s**	Solution $EMF_{ind} = B\ L\ V \sin(\theta)$ $(0.3) = (0.1)(3)(V) \sin(90)$ $(v) = (0.3) / (0.1)(3)$ $(v) = 1$ m/sec
4	The opposite graph represents the relation between the instantaneous electromotive force (emf) produced from an AC dynamo during a complete cycle and time (t). If the face area of the dynamo's coil is $\frac{4}{\pi}$m^2 and the number of its turns is 250 turns, then the dynamo's coil rotates in a magnetic flux density that equals ...... (A) 3.8×10^{-3}T (B) 1.2×10^{-3}T (C) 2.6×10^{-3}T (D) 4.2×10^{-3}T  **Answer:** **(B) 1.2×10^{-3}T**	Solution $emf_{Max} = B\ A\ N\ (2\pi f)$ $(120) = (B)(4/\pi)(250)(2\pi\ 50)$ $(B) = (120) / (4/\pi)(250)(2\pi\ 50)$ $(B) = 0.0012$ T
5	In an AC generator, a rectangular coil of cross-sectional area of 0.07m^2 and 100 turns rotates at a rate of 600 revolution per minute in uniform magnetic flux of density 0.1T. a. Calculate the electromotive force induced in the coil after 0.025s from the position being perpendicular to the direction of the magnetic flux. b. Calculate the electromotive force induced in the coil after 0.05s from the position being perpendicular to the direction of the magnetic flux.	Part a $EMF = BAN(2\pi f)sin\theta$ $sin\theta = \sin sin\ (2 * 180 * f * t)\ =$ $EMF = 0.1 * 0.07 * 100(2\pi * 10$ $emf = 14\pi V = 44V$ **Part b** Emf = 0V
6	State the 4 main components of the AC generator	1. Strong field magnet (permanent magnet or an electromagnet) 2. An armature 3. Two slip rings 4. Two graphite brushes
7	Difference between Fleming right hand and Fleming left hand rule	Fleming right hand rule Used to determine the direction of induced current in a wire moved perpendicular to magnetic flux density

		While Fleming left hand rule Used to determine the direction of the magnetic force affected on wire carrying current putted perpendicular to magnetic flux
8	*A coil of a simple AC generator consists of 100 turns, the cross-sectional area of each turn is 0.21 m^2. The coil rotates with frequency of 50 Hz (cycles/second) in a magnetic field of constant flux density $B = 10^{-3}$ Wb/m^2. What is the maximum generated induced emf? Then calculate the value of the induced emf when the angle between the direction of the linear velocity of the two longitudinal sides of the coil and the flux direction is 30°?*	$(emf)_{max} = NBA\omega = NBA(2\pi f)$ $= 100 \times 10^{-3} \times 0.21 \times 2 \times \frac{22}{7} \times 50 = 6.6\ V$ $emf = (emf)_{max} \sin\theta = 6.6 \times \sin 30 = 6.6 \times \frac{1}{2} = 3.3\ V$
9	Mention the application of 1- eddy current. 2- filming right hand rule.	1-Applications of eddy currents: a) Magnetic braking b) Induction heating 2-Fleming's right-hand rule is used to determine the direction of the induced current in a conductor moving in a magnetic field.
10	When the induced emf in straight wire moved in magnetic field equal to zero?	🎬 The wire is moving parallel to the magnetic field lines. 🎬 The magnetic flux cutting the wire is not changing.
11	When the induced emf in straight wire moved in magnetic field equal to maximum value?	🎬 The wire is moving perpendicular to the magnetic field lines. 🎬 The magnetic flux cutting the wire is changing at the maximum rate.
12	Mention the rule or method used to determine the direction of induced current in moving wire in a uniform magnetic field. (تجريبي)	Fleming's right-hand rule
13	Mention the application of 1- electric generator. 2- Strong field magnet with concave poles in (electric generator). 3- Two slip rings in (electric generator). 4- An armature (coil) in (electric generator). 5- Two graphite brushes in (electric generator).	1- power plants: Generating electricity for homes, businesses, and industries. 2- Strong field magnet with concave poles: Creates a strong and uniform magnetic field 3- Provide electrical contact between the rotating armature and the external circuit.

		4- A loop of wire that rotates in the magnetic field, inducing an emf. 5- The induced currents in the coil pass to the external circuit through them
14	Mention the scientific base of electric generator.	The principle of electromagnetic induction

7. Compare between each of the following

<table>
<tr><th>No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1</td><td colspan="2">Compare
<table>
<tr><td>Point of comparison</td><td>direct current</td><td>alternative current</td></tr>
<tr><td>Direction (2023 دور أول) (تجريبي)</td><td>one direction</td><td>Opposite direction</td></tr>
<tr><td>Source</td><td>It is produced by battery or DC dynamo</td><td>It is produced by AC dynamo</td></tr>
<tr><td>Value</td><td>It has constant intensity and constant direction</td><td>It changes its direction and its intensity periodically</td></tr>
<tr><td>Frequency</td><td>Its frequency = zero</td><td>It has frequency</td></tr>
<tr><td>kind of produced magnetic field</td><td>It produces constant (steady) magnetic field</td><td>It produces variable magnetic field</td></tr>
</table></td></tr>
<tr><td>2</td><td colspan="2">(تجريبي)
<table>
<tr><td>Point of comparison</td><td>The electric generator</td><td>The electric motor</td></tr>
<tr><td>The scientific base</td><td>Faraday's Law of Electromagnetic Induction</td><td>Electromagnetism</td></tr>
</table></td></tr>
</table>

Lesson (5): The electric transformer

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	Device to step-up or step-down voltage	The electric transformer
2	a Device to transform mechanical energy into electrical energy	AC generator (Dynamo)
3	In the transformer the ratio of output electric power to the input electric power. (دور أول 2024) (دور أول / ثان 2023) (تجريبي)	Efficiency of a transformer

2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	the induced voltage produced from the electric transformer. (تجريبي)	Number of Turns in the Primary and Secondary Coils
2		

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	If rotational velocity of a dynamo armature is doubled, then induced e.m.f. will become	a) Half b) Two times c) 4 times d) None	b
2	For a step-down transformer, the number of turns is greater in	a) primary coil b) secondary coil c) both	a
3	For a step-down transformer, the volt is greater in	a) primary coil b) secondary coil c) both	a
4	For a step-down transformer, the current is greater in	a) primary coil b) secondary coil c) both	b
5	For a step-down transformer, the K equals to	a) =1 b) <1 c) >1 d) Not = 1	b
6	In ideal transformer, the efficiency η equals to	a) =1 b) <1 c) >1	a

		d) Not = 1	
7	In a not ideal transformer, the efficiency η equals to	a) =1 b) <1 c) >1 d) Not = 1	b
8	In a step-up transformer Vs > VP implying that Is < Ip. Thus a step up transformer, a) increases the voltage by increasing the current b) increases the voltage by decreasing the current c) decreases the voltage by decreasing the current d) decreases the voltage by decreasing the current		b
9	In an ideal step-up transformer, which of the following statements is incorrect (دور أول 2023) (a) The current ratio of secondary to primary is smaller than one. (b) The turn's ratio of primary to secondary is smaller than one. (c) The voltage ratio of primary to secondary is greater than one. (d) The output-to-input electric power ratio is smaller than one.		D
10	The ratio between the number of turns of the secondary coil to that of the primary coil in an ideal transformer is (3/2), so if the output power of the transformer is (p_W), then the input power at the primary coil equals .... (A) (2/3) P_W (B) 1.5 P_W (C) P_W (D) 5 P_W		C
11	The numbers of turns of the coils in an ideal electric transformer are 400 turns and 200 turns, so if it is connected to an AC source of electromotive force 50V, then the maximum electromotive force that can be obtained is ...... (A) 90V (B) 100V (C) 80V (D) 60V		B

12	An electric transformer of efficiency 90% works on an AC voltage of a maximum value of 141.4V where the consumed power in its primary coil is 3kW and a current intensity of 6A passes in its secondary coil, so the potential difference between the terminals of its secondary coil and the current intensity that passes through its primary coil are …………… (A) (V_s = 400V), (I_p = 30A) (B) (V_s = 400V), (I_p = 15A) (C) (V_s = 450V), (I_p = 15A) (D) (V_s = 450V), (I_p = 30A)	C

4. Correct the underlined in the following statement

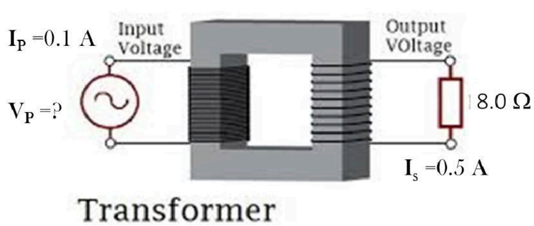
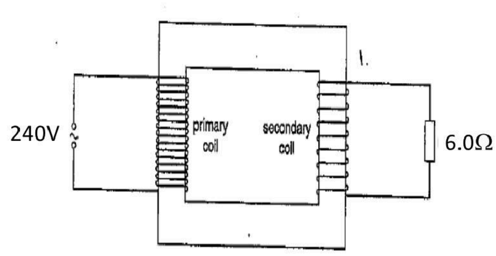
No.	Question	Answer
1	In transformer, core is made of soft iron to reduce force opposing electric current	Eddy current losses
2	A laminated iron core is used in an electric transformer to increase eddy currents. (2024 دور أول)	reduce

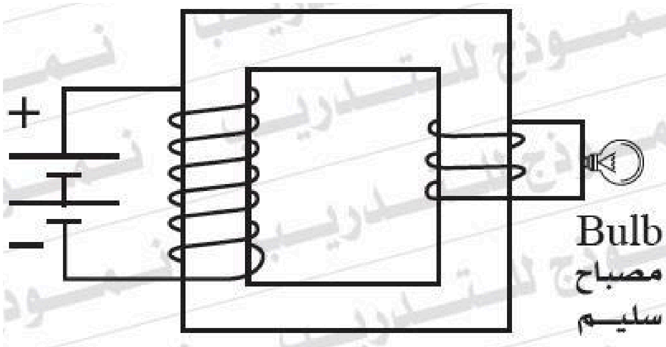
5. True or False

No.	Question	Answer
1	Electric transformer operates for both AC and DC currents	False
2	Electric transformer operates for DC current only.	False
3	Electric transformer operates for AC current only	True
4	Eddy currents are associated with heating effects	True
5	An AC generator can operate without a magnet	False
6	An AC generator is used to convert electrical power into mechanical power	False
7	An AC generator is used to convert mechanical power into electrical power	True
8	DC currents has zero frequency	True

9	AC currents have constant value with time	False
10	DC currents produces constant magnetic field	True

6. Answer the following question

No.	Question	Answer
1	*An ideal step-down transformer is used to operate an electric lamp of power 24 W and potential difference 12 V using AC source of emf 240 V, if the number of turns of the secondary coil is 480 turns, calculate :* (a) The electric current intensity passing through the secondary coil. (b) The number of turns of the primary coil.	(a) $P_w = V_s I_s = 24\ W$ $I_s = \frac{P_w}{V_s} = \frac{24}{12} = 2\ A$ (b) $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $\frac{12}{240} = \frac{480}{N_p}$ $N_p = \frac{480 \times 240}{12} = 9600$ turns
2	The secondary coil of an ideal transformer delivers an r.m.s. current of 0.5 A to a load resistor of resistance 8.0 Ω. The r.m.s. current in the primary is 0.1 A. What is the r.m.s. potential difference across the primary coil? 	20 V
3	The diagram show an iron-cored transformer assumed to be 100% efficient. The ratio of the secondary turns to the primary turns is 1:17. A 240V, a.c. supply is connected to the primary coil and a 6.0Ω resistor is connected to the secondary coil. What is the current in the primary? 	0.14 A

<table>
<tr><td>4</td><td>An ideal step-up transformer, the ratio between the number of its turns is 7:3. Its secondary coil is connected to a device which works on a potential difference 10V. What is the value of primary coil’s volt?</td><td>$\frac{V_p}{V_s} = \frac{N_p}{N_s}$
$\frac{V_p}{(10)} = \frac{3}{7}$
$V_p = \frac{30}{7} = 4.29V$</td></tr>
<tr><td>5</td><td>Why does not the bulb glow?


Electric transformer</td><td>no Ac source connected to the primary coil.</td></tr>
<tr><td>6</td><td>A transformer connected to 220-volt line shows an output of 2 A at 11000 volts. The efficiency is 100%. The current drawn from the line is</td><td>100 A</td></tr>
<tr><td>7</td><td>In a transformer the primary has 500 turns and secondary has 50 turns. 100 volts are applied to the primary coil, the voltage developed in the secondary will be</td><td>10 V</td></tr>
<tr><td>8</td><td>The ratio between the number of turns of the secondary coil to that of the primary coil in an ideal transformer is 4/2, so if the output power of the transformer is 20 watts, then the input power at the primary coil equals to …</td><td>20</td></tr>
<tr><td>9</td><td>An ideal step-down transformer is used to operate an electric lamp, which works on a potential difference of 30 volts. If an electric source of 240 volts is used & the number of tums of the primary coil is 480 turns, calculate the number of tums of the secondary coils. (دور ثان 2023)</td><td>Np / Ns = Vp / Vs
480 turns / Ns = 240 V / 30 V
480 / Ns = 8
Ns = 480 / 8 = 60 turns</td></tr>
<tr><td>10</td><td>in an ideal transformer, the secondary coil has twice as many turns as the primary. If the alternating current in the primary coil is 2A. calculate the current in the secondary coil. (تجريبي)</td><td>$N_p / N_s = I_s / I_p$
Np / Ns = 1/2
1/2 = Is / 2A
I_s = 1A</td></tr>
</table>

7. Compare between each of the following

<table>
<tr><th>No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1</td><td colspan="2">The
<table><tr><td>Point of comparison</td><td>step up transformer</td><td>step down transformer</td></tr><tr><td>transformer ratio value</td><td>K > 1</td><td>K < 1</td></tr></table></td></tr>
</table>

	Use	step up transformer increases the voltage by decreasing the current	step down transformer decreases the voltage by increasing the current